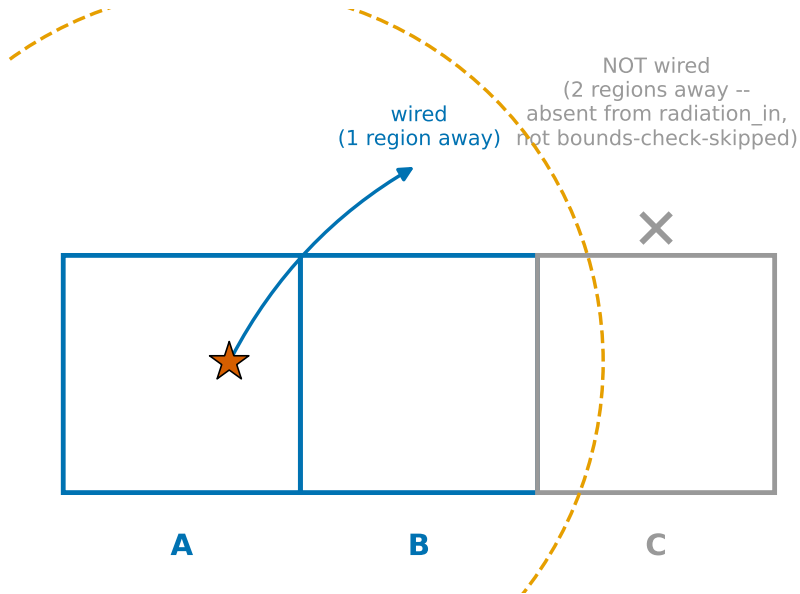


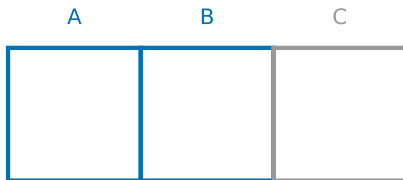
Fig. 5 -- Case 2: radiation_level at leaf level, C two regions away
radius sliver-covers C, but C is structurally invisible to the star's traversal



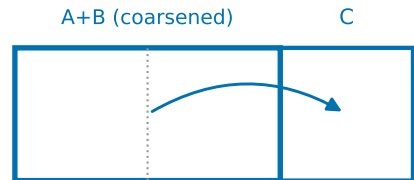
Three guards bound this gap:

1. Split gate -- a region stops splitting once factor $\cdot \text{kernel_gamma} \cdot h_{\text{hij_max}} \geq 0.5 \cdot d_{\text{min}}$
2. Rebuild criterion -- whole-tree rebuild fires once reach + drift $> d_{\text{min}}$
3. In-pass clamp -- dynamic_search_radius capped at interaction_limit + dmin this pass; the outer shell is deferred to the next rebuild, never lost

before rebuild



after rebuild



Rebuild coarsens radiation_level: A and B merge into one region whose 26-neighbour stencil now reaches C.